FNCE 3030: Midterm Exam Study Guide

[1] Money and Capital Markets (Lecture 1, Lecture 1 Supplement, A1)

- Instruments that trade on the money markets and the capital markets, and the difference between exchanges and OTC markets
- Benchmarks used for money market (e.g., SOFR)
- Benchmarks used for the US stock markets (e.g., S&P 500)
- Price-weighted and value-weighted stock market indexes
  - Calculate the value of an index at a given point in time (e.g., A1.P28)
  - Calculate the new divisor when the composition of the index changes (e.g., A1.P31)

[2] Financial Intermediaries and Trading (Lecture 2, Lecture 2 Supplement, A2, PQ1, Q1)

- Types of financial intermediaries
- Mutual funds
  - Calculate a fund's NAV (e.g., A2.P5)
  - Calculate a fun's NAV after redemptions (e.g., A2.P7)
  - Calculate fund returns after fees (e.g., A2.P12)
  - Calculate a fund's taxes on capital gains (e.g., A2.P8)
- Buying on margin (e.g., A2.P28)
  - Calculate account equity and equity percentage (also known as percentage margin)
  - Calculate the principal and the interest on the margin loan
  - Calculate the stock price that would trigger a margin call (e.g., A2.P28 Part 4)
- Short-selling (e.g., A2.P37)
  - Calculate account assets and liabilities
  - Calculate account equity and equity percentage (aka percentage margin)
  - Calculate the stock price that would trigger a margin call (e.g., A2.P37 Part 3)

[3] Return distributions and statistics (Lecture 3, A3, PQ2, Q2)

- Know what risk-free interest rates are and how we calculate them (e.g., T-Bills)
- Know what nominal, real and inflation rates mean
- Calculate a stock's returns from prices and dividends (e.g., A3.P2)
- When historical data is provided:
  - Use Excel to calculate mean return, return volatility and return variance (e.g., A3.P13)
  - Use Excel to calculate percentiles and value-at-risk
  - Use Excel to calculate covariance and correlations between two stocks (e.g., A3.P28)
- When the return distribution is assumed to be a normal distribution:
  - Use Excel to calculate the probability that a stock return is below, above or between two values (e.g., A3.P18)
  - Use Excel to calculate percentiles and value-at-risk (e.g., A3.P22)
- When return values and their probabilities are provided (e.g., A3.P5):
  - Calculate expected return, return volatility and return variance
- Calculate an asset's risk premium and Sharpe's ratio (e.g., A3.P16)

[4] Portfolio construction/analysis (Lecture 4, Lecture 4 Supplement, A3, A4, PQ2, Q2)

- Calculate portfolio weights from dollar values of positions (e.g., A3.P23)
- When historical data is provided:
  - Use Excel to calculate the historical returns of a portfolio from the historical returns of the assets in the portfolio, if the portfolio is rebalanced every month (e.g., A3.P29)
  - Use Excel to calculate a portfolio's holding period return, if the portfolio is never rebalanced (e.g., A3.P30)
  - Use Excel to calculate a portfolio's mean return, return volatility and return variance (e.g., A3.P34)
  - Use Excel to calculate a portfolio's risk premium and Sharpe's ratio (e.g., A3.P37)

- When assets' mean returns, standard deviations, and covariances are provided:
  - Use the portfolio expected return formula to calculate a portfolio's expected return (e.g., A3.P25)
  - Use the portfolio variance formula for 3 assets to calculate a portfolio's return variance and volatility (e.g., A3.P25, A4.P13)
  - Calculate a portfolio's risk premium and Sharpe's ratio

[5] Optimal Portfolio Allocation (Lecture 4, Lecture 4 Supplement, A3, A4, PQ2, Q2)

- Calculate the utility function for a portfolio (A3.P40)
- Use Solver to find the minimum variance portfolio of a set of risky assets (e.g., A4.P3):
  - Set it up: Start with a portfolio that has equal weights across the assets
  - Initial calculations: Calculate the portfolio's expected return and variance
  - Objective: Ask Solver to Minimize the portfolio's variance
  - Changing variable: Ask Solver to change the weights of the assets
  - Constraint: Weights must add up to 1
  - Results: The weights that Solver found are the weights of the minimum variance portfolio
- Use Solver to find the optimal risky portfolio (ORP) of a set of risky assets (e.g., A4.P18-27):
  - Set it up: Start with a portfolio that has equal weights across the assets
  - Initial calculations: Calculate the portfolio's expected return, volatility, and Sharpe's ratio
  - Objective: Ask Solver to Maximize the portfolio's Sharpe's ratio
  - Changing variable: Ask Solver to change the weights of the assets
  - Constraint: Weights must add up to 1
  - Results: The weights that Solver found are the weights of the optimal risky portfolio
- Find the optimal (complete) portfolio of a risky asset and the risk-free asset (e.g., A3.P39):
  - In some problems the risky asset is given, while in other problems the risky asset is the ORP, calculated above

- The portfolios that combine the risk-free asset and the ORP lie on a line known as the *efficient frontier*
- The optimal (complete) portfolio for an investor depends on the investor's risk aversion coefficient, A, and can be calculated with the formula on Lecture 4 slide 25